

Abiram Srikanthasegaram

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PROFESSIONAL SUMMARY

Computer Systems Engineering undergraduate at SLIIT (2023–2027) specialising in bridging hardware and software — from register-level AVR Assembly firmware and closed-loop PID motor control, to custom PCB fabrication and fixed-wing aircraft prototyping. Proven ability to take systems from concept to tested hardware: six complete projects delivered end-to-end, each validated against measurable performance targets. Passionate about applying rapid prototyping, control theory, and systems-level thinking to solve real engineering challenges in embedded systems, firmware, and IoT.

KEY SKILLS

Embedded Firmware	AVR Assembly · C/C++ · Arduino · ATmega328P · Register-level I/O, PWM, ADC, Timer configuration
Control Systems	PID / PI control · MATLAB · Simulink · Closed-loop design · Encoder feedback · Motor control
Hardware Design	PCB Design (KiCad, EasyEDA) 3D CAD & Enclosure Design (Autodesk Fusion 360) Two-layer board fabrication Soldering DRC Signal integrity
Platforms & SoC	Raspberry Pi 4 · ESP32 · Arduino Uno · L298N motor driver · IR sensors & receivers
Software & Tools	Python · Git · Linux · GPIO programming · Real-time GUI (Tkinter) · CSV data logging
Aeronautics	Fixed-wing airframe design · Airfoil & CG analysis · Thrust-to-weight optimisation · XFLR5 · OpenVSP · NASA Glenn simulations
Electronics	Digital logic · Analog circuits · BJT biasing · RC filters · Oscilloscope · Decoupling & noise filtering

ACADEMIC PROJECTS

PID-Controlled DC Motor Stabilisation System

Year 3, Sem 1 – 2026

Raspberry Pi 4 · Python · L298N H-Bridge · Optical Encoder · MATLAB/Simulink · EasyEDA · Two-Layer PCB

- Designed and tuned a closed-loop PI controller ($K_p=1$, $K_i=0.45$) that brought a 12V DC motor from rest to setpoint with overshoot below 15% and steady-state error under 2%, fully satisfying the design specification — validated in MATLAB/Simulink simulation before committing to hardware.
- Fabricated a custom two-layer PCB to replace breadboard wiring, isolating encoder signal paths from high-current motor traces — this directly eliminated the intermittent encoder noise and voltage-drop instability that caused controller divergence during prototype testing.
- Built a real-time Python monitoring application featuring live RPM graphing, dynamic K_p/K_i adjustment sliders, PWM duty cycle display, and automated CSV logging — enabling iterative tuning without reflashing firmware.
- Made a deliberate engineering decision to omit the derivative term: experimental testing confirmed K_d amplified encoder quantisation noise, introducing control output jitter without improving rise time or reducing overshoot — a PI-only design was justified and documented.
- Implemented interrupt-driven encoder pulse counting on Raspberry Pi GPIO with separate power rails for logic and motor circuits, preventing motor switching transients from corrupting speed measurements.

Line-Following Arduino Car with Autonomous Parking

Year 3, Sem 1 – 2026

AVR Assembly · ATmega328P · Arduino Uno · L298N · 3× IR Line Sensors · 38 kHz IR Receiver

- Programmed a fully autonomous mobile robot entirely in AVR Assembly with zero library dependencies — all peripheral configuration (DDR/PORT/PIN registers, Timer0 Fast PWM, ADC) written at register level, demonstrating deep understanding of the ATmega328P architecture.
- Designed an 8-state finite state machine (Initialization → Line Following → Marker Detection → Slow Mode → Parking Detection → Parking Execution → Lost Line Recovery → Stop) that handled every operational scenario tested, achieving a 100% pass rate across all 5 test scenarios.

- Engineered analog IR parking detection using ADC threshold comparison on channel A0, triggering a deterministic 7-step parking manoeuvre sequence — chosen over digital detection specifically to provide a configurable sensitivity margin against ambient IR interference.
- Implemented a lost-line recovery routine performing alternating left/right swing searches with bounded retry counts — preventing the robot from driving indefinitely off-track during sharp turns or surface glare events.
- Achieved stable line tracking through straight paths, left/right gradual curves, sharp turns, and line-loss conditions, with correct parking execution — all tested and passed without a single library call.

Cache Design & Performance Analysis Simulator

Year 2 – 2025

Python · Computer Architecture · Trace-Driven Simulation · AMAT Analysis

- Built a configurable trace-driven cache simulator from scratch in Python, sweeping 36 parameter combinations across line sizes (16B, 32B, 64B, 128B), associativity (1-way, 2-way, 4-way), and replacement policies (LRU, FIFO, Random) — grounded in six testable hypotheses derived from IEEE/ACM literature review.
- Identified 64B as the optimal cache line size: simulation showed miss ratio improving from 0.237 at 16B to 0.155 at 64B — a 35% reduction — before degrading at 128B due to over-fetch and cache pollution, directly confirming the spatial locality sweet-spot hypothesis.
- Demonstrated diminishing returns in associativity: DM-to-2-way produced a meaningful drop in conflict misses, while 2-way-to-4-way yielded negligible further gain — validating the engineering case for 2-way as the optimal cost-performance point for an L1 cache.
- Quantified that LRU, FIFO, and Random replacement policies converged to nearly identical miss ratios (~0.1806) on the test trace at steady state — leading to the conclusion that policy selection is workload-dependent and that over-engineering replacement logic is unwarranted for workloads with low capacity pressure.
- Determined the optimal L1 configuration of 64B lines, 2-way associativity, LRU policy achieving an AMAT of ~13.0 ns — a result that aligns with industry-standard L1 design practice and was used as the basis for multi-level hierarchy design recommendations.

CLIMS – Smart Surveillance & Access Control System

Year 1 – 2024

ESP32 · IoT · Python · OpenCV Facial Recognition · Electronic Door Lock · Mobile Push Notifications

- Designed and built a complete end-to-end IoT security pipeline on ESP32: PIR motion trigger → camera capture → Python facial recognition → electronic lock actuation → real-time mobile push notification, with the full decision workflow executing on a single embedded platform.
- Engineered the system to distinguish between known and unknown faces before triggering the lock, preventing false positives from ambient motion — the only action that fires an alert is a confirmed unrecognised face, not general movement.

1-Digit Asynchronous Up Counter – PCB Design & Fabrication

Year 2 – 2025

EasyEDA · 74HC74 D Flip-Flops · Seven-Segment Display · PCB Fabrication · Oscilloscope Validation

- Completed the full hardware lifecycle for a 0–9 asynchronous decimal counter — EasyEDA schematic capture → multi-layer PCB layout → DRC sign-off → board fabrication → component soldering → oscilloscope validation of switching waveforms at each flip-flop stage.
- Verified correct decade counting behaviour and clean logic transitions on oscilloscope, confirming that the physical board matched the simulated timing behaviour — providing hands-on experience with the gap between simulation and fabricated hardware.

PERSONAL PROJECTS

Agni Wing – Fixed-Wing Aircraft Prototype

2025 – Ongoing

Airframe Design · A2212 1125KV BLDC · 40A ESC · 3S 2200mAh LiPo · 1045/1047 Propeller · CAD · NASA Glenn Aeronautics Simulations

- Co-founded a four-person self-directed engineering team (Agni Wing) with the long-term goal of building an autonomous fixed-wing UAV — beginning with a manually controlled prototype to validate aerodynamic fundamentals before integrating flight control electronics.
- Completed the full airframe design cycle within 6 days: aerodynamic layout, fuselage sizing, CG calculation, propeller offset, thrust-line alignment, and wing-fuselage integration — using NASA Glenn's Beginner's Guide to Aeronautics interactive simulations for airfoil and chord analysis, and CAD for dimensional definition.
- Ran 5 iterative build-test-crash-analyse-rebuild cycles — the aircraft achieved approximately 12 seconds of sustained flight reaching 30–40 ft altitude on Attempt 5, validating that the CG positioning and thrust-to-weight ratio (AUW ~850–900 g) were within flyable margins.

- Each failed attempt produced a structured failure analysis: Attempts 1–3 identified CG instability and insufficient thrust margin; Attempt 4 revealed structural weakness at the wing root under load; Attempt 5 incorporated all corrections and achieved the first controlled flight.
- Engineering learnings applied to next phase: aerodynamic validation using XFLR5/OpenVSP, improved structural design for wing-root joint, extended controlled flight envelope, and groundwork for autonomous flight control integration.

CERTIFICATIONS & TRAINING

PLC & Industrial Automation Workshop (PLCFI)

2025

IEEE Student Branch, Curtin University Colombo · Sri Lanka Institute of Robotics (SLIR)

- Completed hands-on sessions covering PLC fundamentals, ladder logic programming, HMI/SCADA configuration using XINJE PLC hardware, and live motor and sensor control demonstrations. Awarded certificate of completion by SLIR.

EDUCATION

BSc (Hons) Computer Systems Engineering

Nov 2023 – Nov 2027 (Expected)

Sri Lanka Institute of Information Technology (SLIIT)

- Relevant modules: Embedded Systems Design · Control System Engineering · Digital Electronics · Computer Architecture · Analog Electronics · Systems Modelling & Prototyping

REFERENCES

Two confirmed referees — full contact details available upon request.

Mr. Dinith Primal Head of Department – Computer Systems Engineering MSc Wireless Communications Sri Lanka Institute of Information Technology (SLIIT) <i>Relationship: Head of Department — confirmed as referee</i>	Mr. Sanjeeva Perera Senior Academic Fellow – Faculty of Computing BSc Eng (University of Moratuwa) · MBA (University of Colombo) Sri Lanka Institute of Information Technology (SLIIT) <i>Relationship: IE3014 Professional Skills lecturer — confirmed as referee</i>
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